

Paper 1B drill: uncertainties, set 2

Putting the rules together, and judging a result.

IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **26 marks**

Set 1 drilled the three rules one at a time. Here they arrive together, and then the harder question: once you have a result with an uncertainty, what may you actually conclude from it?

A *several quantities at once*

- A1. The density of a metal is found from $\rho = m/V$, with $m = 154 \pm 1$ g and $V = 21.5 \pm 0.5$ cm³. Determine ρ with its absolute uncertainty. [3]
- A2. A cylinder has radius 1.20 ± 0.02 cm and height 5.40 ± 0.05 cm. Determine its volume $V = \pi r^2 h$ with its absolute uncertainty. [3]
- A3. A pendulum gives $g = \frac{4\pi^2 L}{T^2}$, with $L = 0.850 \pm 0.005$ m and $T = 1.85 \pm 0.02$ s. Determine g with its absolute uncertainty, and state which measurement limits the result. [4]

B *the uncertainty in a gradient*

- B1. A student's steepest line has gradient 2.34 and the shallowest has 1.98. State the gradient with its uncertainty, and its percentage uncertainty. [3]
- B2. For a pendulum, the gradient of T^2 against L equals $4\pi^2/g$. A student measures the gradient as 4.05 ± 0.15 s² m⁻¹. Determine g with its absolute uncertainty. [3]

C *two different percentages*

- C1. An experiment gives $g = 9.4 \pm 0.5$ m s⁻²; the accepted value is 9.81 m s⁻². Find the percentage *difference* from the accepted value and the percentage *uncertainty* in the result, then state whether the experiment is consistent with the accepted value. [3]
- C2. Another experiment gives 2.55 ± 0.05 where theory predicts 2.72. Answer the same three questions. [3]
- C3. Explain, using your two answers, why a small percentage difference is not the same thing as a good result. [2]

D *do two results agree?*

- D1. Group A measures the speed of sound as 336 ± 8 m s⁻¹; group B gets 349 ± 6 m s⁻¹. State whether the two results agree, with a reason. [2]

- D2. Group C gets $331 \pm 3 \text{ m s}^{-1}$ and group D gets $344 \pm 4 \text{ m s}^{-1}$. State whether these agree, and what it suggests if they do not. **[3]**
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E *working backwards*

- E1. A specific heat capacity is to be found to within 5%. The terms other than the temperature rise already contribute 2.6% in total. The rise is about 14.2°C . Determine the largest absolute uncertainty in the rise that is acceptable. **[3]**
- E2. A length of about 2.0 cm is to be measured to within 1%. Determine the absolute uncertainty required, and state whether a millimetre ruler is good enough. **[3]**

Answers

A *several quantities at once*

- A1. $\rho = 154/21.5 = 7.16 \text{ g cm}^{-3}$. Percentages: $1/154 = 0.65\%$ and $0.5/21.5 = 2.33\%$, adding to 2.97% . Then 2.97% of 7.16 is 0.21 , so $\rho = 7.16 \pm 0.21 \text{ g cm}^{-3}$.
- A2. $V = \pi(1.20)^2(5.40) = 24.4 \text{ cm}^3$. The radius is *squared*, so its percentage is doubled: $2 \times (0.02/1.20) = 2 \times 1.67\% = 3.33\%$, plus $0.05/5.40 = 0.93\%$, giving 4.26% . Then 4.26% of 24.4 is 1.0 , so $V = 24.4 \pm 1.0 \text{ cm}^3$.
- A3. $g = \frac{4\pi^2(0.850)}{(1.85)^2} = 9.80 \text{ m s}^{-2}$. Percentages: L gives $0.005/0.850 = 0.59\%$; T is squared so it gives $2 \times (0.02/1.85) = 2.16\%$. Total 2.75% , and 2.75% of 9.80 is 0.27 , so $g = 9.80 \pm 0.27 \text{ m s}^{-2}$. The *period* limits the result: it contributes 2.16 of the 2.75 percentage points, nearly four times as much as the length, precisely because it is squared.
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B *the uncertainty in a gradient*

- B1. The best value is the midpoint, $\frac{1}{2}(2.34 + 1.98) = 2.16$, and the uncertainty is half the spread, $\frac{1}{2}(2.34 - 1.98) = 0.18$. So 2.16 ± 0.18 , a percentage uncertainty of $0.18/2.16 = 8.3\%$.
- B2. $g = \frac{4\pi^2}{\text{gradient}} = \frac{4\pi^2}{4.05} = 9.75 \text{ m s}^{-2}$. Since g is inversely proportional to the gradient, its percentage uncertainty is the *same* as the gradient's: $0.15/4.05 = 3.7\%$. Then 3.7% of 9.75 is 0.36 , so $g = 9.7 \pm 0.4 \text{ m s}^{-2}$. (Taking a reciprocal does not change a percentage uncertainty; it is a power of -1 , and the magnitude of the power is 1 .)
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C *two different percentages*

- C1. Percentage difference = $\frac{|9.4 - 9.81|}{9.81} \times 100 = 4.2\%$. Percentage uncertainty = $0.5/9.4 = 5.3\%$. The result spans 8.9 to 9.9 , which contains 9.81 , so the experiment *is* consistent with the accepted value.
- C2. Percentage difference = $\frac{|2.55 - 2.72|}{2.72} \times 100 = 6.3\%$. Percentage uncertainty = $0.05/2.55 = 2.0\%$. The result spans 2.50 to 2.60 , which does *not* contain 2.72 , so this experiment is **not** consistent with the prediction. Something systematic is unaccounted for.
- C3. The two percentages answer different questions. The percentage *difference* says how far your value sits from the expected one; the percentage *uncertainty* says how far it was entitled to sit. Only the comparison between them means anything. C1 is further from the accepted value in percentage terms, yet it agrees, because its uncertainty is larger than its discrepancy. C2 is closer and yet disagrees, because its discrepancy is three times its uncertainty. A precise result that misses is a more serious problem than a vague one that happens to land nearby, because precision removes the excuse.

D *do two results agree?*

- D1. Group A spans 328 to 344; group B spans 343 to 355. These overlap between 343 and 344, so the results *do* agree, though only barely. Two results agree when their ranges share any value at all, since that value could be the true one for both.
- D2. Group C spans 328 to 334 and group D spans 340 to 348. There is no overlap, so the results genuinely disagree. At least one group has a systematic error that its stated uncertainty does not cover: something is wrong with a method or an instrument, not merely with the scatter. Quoting a tight uncertainty does not make a result right, and here the tightness is what makes the disagreement undeniable.

E *working backwards*

- E1. The percentages add, so the temperature-rise term may contribute at most $5.0 - 2.6 = 2.4\%$. Of a rise of 14.2°C , that is $0.024 \times 14.2 = 0.34^\circ\text{C}$. Since the rise is a difference of two readings, each thermometer reading would need an uncertainty of about 0.17°C , so a thermometer resolving to 0.1°C is required. A 1°C scale cannot do it.
- E2. 1% of 2.0 cm is 0.02 cm, that is 0.2 mm. A millimetre ruler gives about ± 1 mm on a length, since both ends must be read, which is 5%: five times too coarse. Use vernier callipers, whose ± 0.02 mm gives 0.1%, comfortably inside the target. This is the calculation to do *before* an experiment, when the choice of instrument is still open.